

3a	upthrust and weight	B1
3b	upthrust greater than weight so resultant force is upwards	B1
3c(i)	A , g and ρ all constant so F is proportional to x minus sign means F and x are in opposite directions	B1 B1
3c(ii)	$a = \frac{F}{m} = (-)\frac{A\rho g}{m}x$ compare with SHM defining equation $a = -\omega^2 x$ $\omega^2 = \frac{A\rho g}{m}$ $\omega = \sqrt{\frac{A\rho g}{m}}$	M1 M1 A0
3d(i)	Damping due to viscous forces	B1
3d(ii)	$-\omega^2 = \text{gradient} = \frac{2.30}{0.020} = 115$ $(2\pi f)^2 = 115$ $f = 1.71 = 1.7 \text{ Hz}$	C1 A1
3d(iii)	$E = \frac{1}{2}kx^2 = \frac{1}{2}m\omega^2 x^2$ $\text{initial } E = \frac{1}{2}(0.57)(115)(0.020)^2 = 0.01311 \text{ J}$ $\text{final } E = \frac{1}{2}(0.57)(115)(0.016)^2 = 0.0083904 \text{ J}$ $\Delta E = \text{Initial } E - \text{final } E$ $= 4.7 \times 10^{-3} \text{ J}$ OR accept if students use area under graph to find initial and final E read off first point (-0.02, 2.3) gives initial E = 0.01311 J accept either: read off second point as (0.016, -1.8) gives final E = 0.008208, hence $\Delta E = 4.9 \times 10^{-3} \text{ J}$ read off second point as (0.016, -1.84) gives final E = 0.0083904 hence $\Delta E = 4.9 \times 10^{-3} \text{ J}$	C1 C1 A1
Total:		12

42	Zero electric field strengths in	P1
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